

HARSHDIP SAHA

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Education

Netaji Subhas University of Technology, New Delhi

B.Tech CSE (AI) · GPA 8.78 of 10 · class of 2027

Kendriya Vidyalaya No. 2, Delhi Cantt

Class XII, 95.6% · Class X, 98.4%

Relevant Coursework

- Computer Vision
- Natural Language Processing
- Game Theory
- Artificial Intelligence
- DBMS
- Machine Learning

Experience

AI Engineer Intern — AI-DLC Pilot Team

Jun – Aug 2026

Optum (UnitedHealth Group)

- Built an internal assistant, in TypeScript on the company's agent framework, that helps up to 35,000 employees find the approved, governed way to use AI for their specific role — instead of searching policy documents.
- Built an automated judge for an internal hackathon: a model that scored, consistently and by fixed rules, how well each team followed AI-DLC (the company's development process, where every change carries a written plan and record), so the human judges had a reliable second opinion.
- Interviewed seven kinds of specialist — up to six people each — and turned what they told me into role-specific workflows, prompts, and guardrails the assistant could hand out.

Research Intern — Scientific Computing (remote)

Jan 2026 – present

IIT Madras

- Developing PyAMorph, a Python library that turns images, CAD models, and equations into geometry a physics simulation can run on. It works by computing, for every point in space, how far it is from the nearest surface (a signed distance function).
- Added shape combination (union, subtraction, intersection), conversion from standard 3D mesh files (STL), and GPU-accelerated bindings, all plugged into AMReX — the US Department of Energy's framework for large simulations on supercomputers.
- The work was selected for an oral presentation at INCAM 2026, India's national applied-mathematics conference, at IIT Kanpur; a journal paper is in preparation.

Amazon ML Summer School

Jul – Aug 2026

Amazon

- Selected for Amazon's competitive machine-learning summer programme for students.

Projects

AToM-Net: Neuro-Symbolic LLM Negotiation Agent | LLMs, NLP, Bayesian Inference, Game Theory Feb 2026

- Designed a **neuro-symbolic NLP architecture** for multi-turn negotiation, combining **LLM-based dialogue understanding** with a **Bayesian belief tracking module** to model opponent intent.
- Introduced an **Economic Consistency Validator** to detect contradictions between natural language ("cheap talk") and actions ("costly signals"), improving reasoning robustness.
- Implemented a **Reflexion-based memory module** enabling inter-dialogue learning without weight updates, improving strategic adaptation across conversations.
- Achieved **23.30 vs 12.53 average score** against baseline LLMs on the CaSiNo negotiation benchmark, demonstrating superior language-grounded decision making.
- First architectural work to present results on the NegotiationTOM dataset, achieving SOTA performance with an overall score of **16.29**. ([Paper](#)).

RECAP-Net: Classification of Active Progression | PyTorch, MONAI, GANs, Swin UNETR, 3D CNNs Sept 2025

- Proposed an end-to-end pipeline for the **BraTS-PRO 2025 Challenge**, performing longitudinal glioblastoma response classification under the **RANO criteria**, integrating **Swin UNETR** segmentation with an ensemble of 3D ResNet, DenseNet, and EfficientNet classifiers.
- Applied **z-score normalization**, multimodal fusion of baseline/follow-up MRIs, and a **3D spectrally normalized GAN** for class balancing, achieving **3D-SSIM: 0.92 ± 0.03** and **KL divergence: 0.037 ± 0.004**.

- Accepted for **oral presentation at MICCAI 2025**, a top-tier (A*) conference in medical image computing.
- Achieved **94.0% balanced accuracy**, **94.6% F1-score**, and **96.0% AUROC**, outperforming individual backbones ([Paper](#)).

Brand-Aware Diffusion for Automated Marketing Poster Generation | *NLP, CLIP, Diffusion Models* Jan 2026

- Developed a **brand-aware generative system** that combines **diffusion models (Stable Diffusion XL)** with **LLM-based text generation** for automated marketing poster creation.
- Designed a pipeline integrating **CLIP-based semantic understanding** and **context-aware NLP generation** for headlines, subheadlines, and CTAs based on user prompts.
- Implemented **context detection (festival, product launch, sales)** using LLM inference, enabling adaptive multimodal generation aligned with brand identity.
- Built a FastAPI backend and Adobe Express add-on enabling **interactive text editing and layer-wise control** over generated outputs.
- This project **won the Adobe** enterprise hackathon. ([Paper](#)).

AI Loan Assistant – WhatsApp & Website Chatbot | *NLP, XGBoost, Twilio API, Firebase, OCR* Dec 2025

- Developed an AI-powered loan origination platform enabling users to apply for loans through both Website Chat and WhatsApp using a centralized multi-agent orchestration system.
- Built automated KYC verification, OCR-based document parsing, and credit risk assessment pipelines using **XGBoost** for real-time loan approval or rejection decisions.
- Integrated **Twilio WhatsApp API**, multilingual support using DeepTranslator, and sanction letter PDF generation with ReportLab for automated delivery via WhatsApp or website download ([Demo Video](#)).

Accurate Precise Timely | *Machine Learning, XGBoost, LSTM, ARIMA, Greedy scheduling* Mar 2025

- Developed an AI-powered system that schedules doctor appointments based on patient condition severity and predicted consultation time, while aligning with doctor availability and shift timings.
- Built an admin panel for real-time hospital resource monitoring and implemented LSTM and ARIMA models to predict next-day needs, refined daily using staff updates ([Demo Video](#)).
- Selected among the **top 30** teams to pitch offline at the **AI4Humanity Summit** (with the Israel Embassy) in a hackathon of 150+ teams ([Certificate](#)).

Technical Skills

Languages: Python, C/C++, [MATLAB](#), SQL, JavaScript, HTML/CSS

Frameworks and Tools: Tensorflow, OpenCV, PyTorch, Pygame, Scikitlearn, pytesseract (OCR), Tkinter, Streamlit, SHAP (xAI), Git, Azure, GCP, spaCy, nltk, LangChain, Jupyter, statsmodels, Visual Studio Code

Databases and Services: Azure Blob Storage, MySQL, PostgreSQL, Hugging Face, [Kaggle](#)

Core Competencies: [Computer Vision](#), ML, Data Analysis, Natural Language Processing, Cloud Computing, Time Series Analysis, Deep Learning, [AI](#)

Key Achievements and Extracurriculars

- Secured **World Rank 3** in the **BraTS Lighthouse 2025 – Tumor Progression Challenge**, with our paper accepted for **Oral Presentation** at the **MICCAI 2025 Conference, South Korea**.
- Currently working under the **NextGen AI Lab, NSUT**, contributing to advanced research including glioblastoma response classification using multimodal MRI data, NLP and deep learning architectures.
- Top open-source contributor to [BrainGlobe](#) ecosystem; merged **15+ pull requests** across repositories and a leading contributor to [brainglobe-registration](#), improving neuroscience image registration pipelines.
- Active ML member at Google Developer Students Club (GDSC) and core member at Hackethix Society.
- Secured All India 14th rank in BRAINDEAD Unstop competition by making two projects: one on EDA and ML, and another on an LLMs hybrid summarizer in Feb 2025. ([GitHub Link](#)) ([Certificate Link](#))
- Solved **1000+** problems across various coding platforms such as [LeetCode](#), [GeeksforGeeks](#), and others. Achieved **Pupil** rank on [Codeforces](#).
- Ranked in the top 0.43% out of 1.1M students in JEE-2023.